Question ID 56e373b3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	■□□

ID: 56e373b3

SAT6 M 3.1

A manager is responsible for ordering supplies for a shaved ice shop. The shop's inventory starts with $4,500$ paper cups, and the manager estimates that 70 of these paper cups are used each day. Based on this estimate, in how many days will the supply of paper cups reach $1,700$?

A. 20

B. 40

C. 60

D. 80

Question ID b434e103

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	■□□

ID: b434e103

SAT6 M 3.2

In $\triangle RST$, the measure of $\angle R$ is 63°. Which of the following could be the measure, in degrees, of $\angle S$?

A. 116

B. 118

C. 126

D. 180

Question ID d9733ed9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	■■□

ID: d9733ed9

SAT6 M 3.3

$y > 4x + 8$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
2	19
4	30
6	41

B.

x	y
2	8
4	16
6	24

C.

x	y
2	13
4	18
6	23

D.

x	y
2	13
4	21
6	29

Question ID 1f0966db

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	■■□

ID: 1f0966db

SAT6 M 3.4

The function f is defined by $f(x)=\frac{9}{7}x+\frac{8}{7}$. For what value of x does $f(x)=5$?

Question ID 842cec4d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	■□□

ID: 842cec4d

SAT6 M 3.5

During a portion of a flight, a small airplane's cruising speed varied between 150 miles per hour and 170 miles per hour. Which inequality best represents this situation, where s is the cruising speed, in miles per hour, during this portion of the flight?

A. $s \leq 20$

B. $s \leq 150$

C. $s \leq 170$

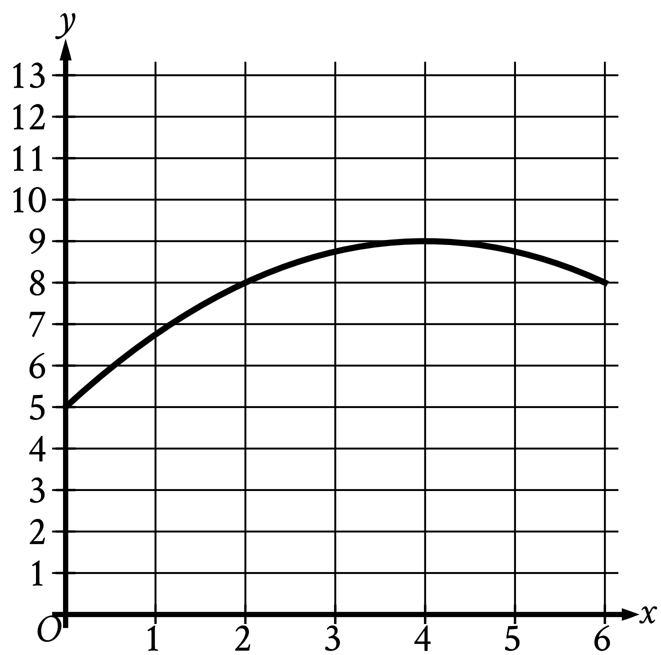
D. $150 \leq s \leq 170$

Question ID 95d1c344

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■■□

ID: 95d1c344

SAT6 M 3.6



The graph models the number of active projects a company was working on x months after the end of November 2012, where $0 \leq x \leq 6$. According to the model, what is the predicted number of active projects the company was working on at the end of November 2012?

A. 0

B. 5

C. 8

D. 9

Question ID 4b0c156b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	■■□

ID: 4b0c156b

SAT6 M 3.7

In the linear function h, $h(28) = 15$ and $h(26) = 22$. Which equation defines h?

A. $h\left(x\right) = -\frac{2}{7}x + 23$

B. $h\left(x\right) = -\frac{2}{7}x + 113$

C. $h\left(x\right) = -\frac{7}{2}x + 23$

D. $h\left(x\right) = -\frac{7}{2}x + 113$

Question ID 9551cd91

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	■■□

ID: 9551cd91

SAT6 M 3.8

Number of cars	Maximum number of passengers and crew
3	174
5	284
10	559

The table shows the linear relationship between the number of cars, c, on a commuter train and the maximum number of passengers and crew, p, that the train can carry. Which equation represents the linear relationship between c and p?

A. $55c - p = -9$

B. $55c - p = 9$

C. $55p - c = -9$

D. $55p - c = 9$

Question ID a267bd29

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■■□

ID: a267bd29

SAT6 M 3.9

$w^2 + 12w - 40 = 0$ Which of the following is a solution to the given equation?

A. $6 - 2\sqrt{19}$

B. $2\sqrt{19}$

C. $\sqrt{19}$

D. $-6 + 2\sqrt{19}$

Question ID 9f70fd47

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	■■■

ID: 9f70fd47

SAT6 M 3.10

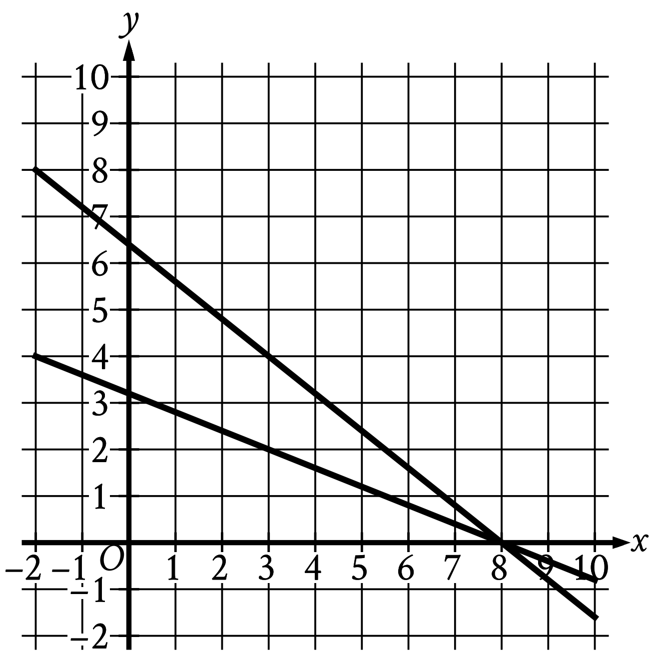
What is the *y*-coordinate of the *y*-intercept of the graph of $\frac{3x}{7} = -\frac{5y}{9} + 21$ in the *xy*-plane?

Question ID 3f5a3602

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	■■■

ID: 3f5a3602

SAT6 M 3.11



What system of linear equations is represented by the lines shown?

A. $8x + 4y = 32$ $-10x - 4y = -64$

B. $8x - 4y = 32$ $-10x + 4y = -64$

C. $4x - 10y = 32$ $-8x + 10y = -64$

D. $4x + 10y = 32$ $-8x - 10y = -64$

Question ID 3a84f885

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	■■■

ID: 3a84f885

SAT6 M 3.12

$(x-2)-4(y+7)=117$ $(x-2)+4(y+7)=442$

The solution to the given system of equations is (x, y). What is the value of $6(x-2)$?

Question ID 358f18bc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■■■

ID: 358f18bc

SAT6 M 3.13

$f(x) = x^2 - 48x + 2{,}304$ What is the minimum value of the given function?

Question ID 0393ba15

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	■■■

ID: 0393ba15

SAT6 M 3.14

$-49x = -98x$ How many solutions does the given equation have?

A. Zero

B. Exactly one

C. Exactly two

D. Infinitely many

Question ID 6e7ae9fc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■■■

ID: 6e7ae9fc

SAT6 M 3.15

The function g is defined by $g(x) = x(x-2)(x+6)^2$. The value of $g(7-w)$ is 0, where w is a constant. What is the sum of all possible values of w?

Question ID e438ec3f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	■■■

ID: e438ec3f

SAT6 M 3.16

A grove has 6 rows of birch trees and 5 rows of maple trees. Each row of birch trees has 8 trees 20 feet or taller and 6 trees shorter than 20 feet. Each row of maple trees has 9 trees 20 feet or taller and 7 trees shorter than 20 feet. A tree from one of these rows will be selected at random. What is the probability of selecting a maple tree, given that the tree is 20 feet or taller?

A. $\frac{9}{164}$

B. $\frac{3}{10}$

C. $\frac{15}{31}$

D. $\frac{9}{17}$

Question ID 44b2b894

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	■■■

ID: 44b2b894

SAT6 M 3.17

A rectangle is inscribed in a circle, such that each vertex of the rectangle lies on the circumference of the circle. The diagonal of the rectangle is twice the length of the shortest side of the rectangle. The area of the rectangle is $1{,}089\sqrt{3}$ square units. What is the length, in units, of the diameter of the circle?

Question ID 4aaa9c42

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	■■■

ID: 4aaa9c42

SAT6 M 3.18

The positive number a is $2{,}241\%$ of the sum of the positive numbers b and c, and b is 83% of c. What percent of b is a?

A. 23.24%

B. 49.41%

C. $2{,}324\%$

D. $4{,}941\%$

Question ID 59352689

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	■■■

ID: 59352689

SAT6 M 3.19

$$\frac{7}{8}y - \frac{5}{8}x = \frac{4}{7} - \frac{7}{8}y$$
$$\frac{5}{4}x + \frac{7}{4} = py + \frac{15}{4}$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p?

Question ID 02060533

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■■■

ID: 02060533

SAT6 M 3.20

x	$g(x)$
-27	3
-9	0
21	5

The table shows three values of x and their corresponding values of $g(x)$, where $g(x) = \frac{f(x)}{x+3}$ and f is a linear function. What is the *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane?

A. $(0, 36)$

B. $(0, 12)$

C. $(0, 4)$

D. $(0, -9)$

Question ID 0acfddb5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	■■■

ID: 0acfddb5

SAT6 M 3.21

A circle has center G, and points M and N lie on the circle. Line segments MH and NH are tangent to the circle at points M and N, respectively. If the radius of the circle is 168 millimeters and the perimeter of quadrilateral $GMHN$ is $3{,}856$ millimeters, what is the distance, in millimeters, between points G and H?

A. 168

B. $1{,}752$

C. $1{,}760$

D. $1{,}768$

Question ID 95eeeb5b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■■■

ID: 95eeeb5b

SAT6 M 3.22

The function f is defined by $f(x) = ax^2 + bx + c$, where a, b, and c are constants. The graph of $y = f(x)$ in the *xy*-plane passes through the points $(7, 0)$ and $(-3, 0)$. If a is an integer greater than 1, which of the following could be the value of $a + b$?

A. -6

B. -3

C. 4

D. 5